

Statistical Analysis Report – Student Performance in Exams (ANOVA)

Overview

This project investigated whether student mathematics performance differs across levels of parental education using the **Students Performance in Exams** dataset obtained from Kaggle. The analysis used descriptive statistics, data visualization, distribution diagnostics, and a one-way Analysis of Variance (ANOVA) to determine whether differences in average mathematics scores existed between parental education groups. In addition, effect size analysis was conducted to assess the practical significance of any observed differences.

Github Repo Link: <https://github.com/Shehzaad-S/udacity-statistical-analysis-project>

Dataset Source:

Students Performance in Exams Dataset (Kaggle)

Dataset Link: <https://www.kaggle.com/datasets/spscientist/students-performance-in-exams>

Dataset Description

The Students Performance in Exams dataset contains academic performance and demographic information for 1,000 students. The dataset includes 8 variables describing student achievement and background characteristics, including math, reading, and writing scores, parental level of education, gender, race/ethnicity, lunch status, and test preparation course participation.

The primary dependent variable examined in this analysis was **math score**, while **parental level of education** served as the grouping variable for the hypothesis test. The parental education variable contains six educational categories ranging from “some high school” to “master's degree” qualifications.

Initial inspection confirmed that the dataset contained no missing values and that all variables were loaded with appropriate data types. As a result, only minimal data cleaning and validation procedures were required prior to analysis.

Methods

Initial Data Analysis

The analysis began with descriptive statistics and exploratory visualizations to understand the characteristics of the dataset and evaluate its suitability for inferential analysis. Lusa et al. (2024) emphasize that Initial Data Analysis (IDA) should include numerical summaries, graphical exploration, assessment of data quality, and evaluation of assumptions relevant to planned statistical analyses. Consistent with these recommendations, numerical summaries, categorical frequency counts, and graphical representations were examined before formal hypothesis testing.

Descriptive statistics were calculated for math, reading, and writing scores to evaluate measures of central tendency and dispersion. Frequency counts were generated for categorical variables to understand the representation of subgroups within the dataset.

Visual Models

Three visual models were created to investigate patterns relevant to the research question:

Histogram of Mathematics Scores:

The histogram was used to examine the shape of the mathematics score distribution, identify potential skewness, detect unusual observations, and assess whether the variable appeared suitable for parametric analysis.

Boxplot of Mathematics Scores by Parental Education:

The boxplot was selected because it provides a visual summary of medians, interquartile ranges, variability, and potential outliers while supporting comparisons across multiple groups. This visualization was particularly useful for comparing mathematics performance across parental education categories.

Mean Mathematics Score by Parental Education Level:

A grouped bar chart with 95% confidence intervals was used to compare average math scores across parental education levels.

The use of both univariate and multivariate graphical summaries is consistent with recommended Initial Data Analysis practices for understanding variable distributions and relationships prior to conducting inferential analyses (Lusa et al., 2024).

Distribution Diagnostics

Additional diagnostics were conducted to supplement visual assessment of the mathematics score distribution. Skewness was used to assess distributional asymmetry, while kurtosis was used to evaluate tail behavior relative to a normal distribution (Agarwal, Prabha, & Yadav, 2025).

The mathematics score distribution exhibited only slight negative skewness and a kurtosis value close to zero, suggesting that the distribution was reasonably similar to a normal distribution. Given the large sample size ($n = 1,000$), these minor deviations from normality were not considered sufficient to invalidate the planned analysis.

Hypothesis Testing

A one-way ANOVA was selected because the analysis aimed to compare mean mathematics scores across six independent parental education groups. Ostertagová and Ostertag (2013), explain that one-way ANOVA is appropriate when evaluating whether means differ across multiple groups defined by a single categorical factor.

The hypotheses were:

Null Hypothesis (H_0):

The mean mathematics score is equal across all parental education groups.

Alternative Hypothesis (H_1):

At least one parental education group has a mean mathematics score that differs from the others.

Assumption Testing

Before conducting the ANOVA, the assumption of homogeneity of variance was evaluated using Levene's Test. Equality of variances is an important assumption for analysis of variance procedures Ostertagová and Ostertag (2013).

Levene's Test produced a p-value of 0.458, indicating insufficient evidence to conclude that group variances differed significantly. This supported the assumption of homogeneity of variance required for the ANOVA.

Results

Descriptive statistics indicated that math, reading, and writing scores exhibited similar average performance levels, with means ranging from approximately 66 to 69 points. Standard deviations were consistent across subjects at approximately 15 points, suggesting moderate variability among students (looking at the means for reference).

The histogram demonstrated a unimodal mathematics score distribution with slight negative skewness. Distribution diagnostics supported the visual assessment, with skewness (-0.279) and kurtosis (0.275) suggesting only minor departures from normality.

The boxplot and grouped mean comparison chart both suggested a positive relationship between parental education and mathematics performance. Students whose parents possessed higher educational attainment generally exhibited higher median and average mathematics scores. However, overlap among confidence intervals indicated that visual inspection alone was insufficient to determine statistical significance.

The one-way ANOVA produced an F-statistic of 6.522 and a p-value of 0.000006. Because the p-value was substantially smaller than the significance level of 0.05, the null hypothesis was rejected. The analysis therefore provides statistically significant evidence that mathematics performance differs across at least one parental education group.

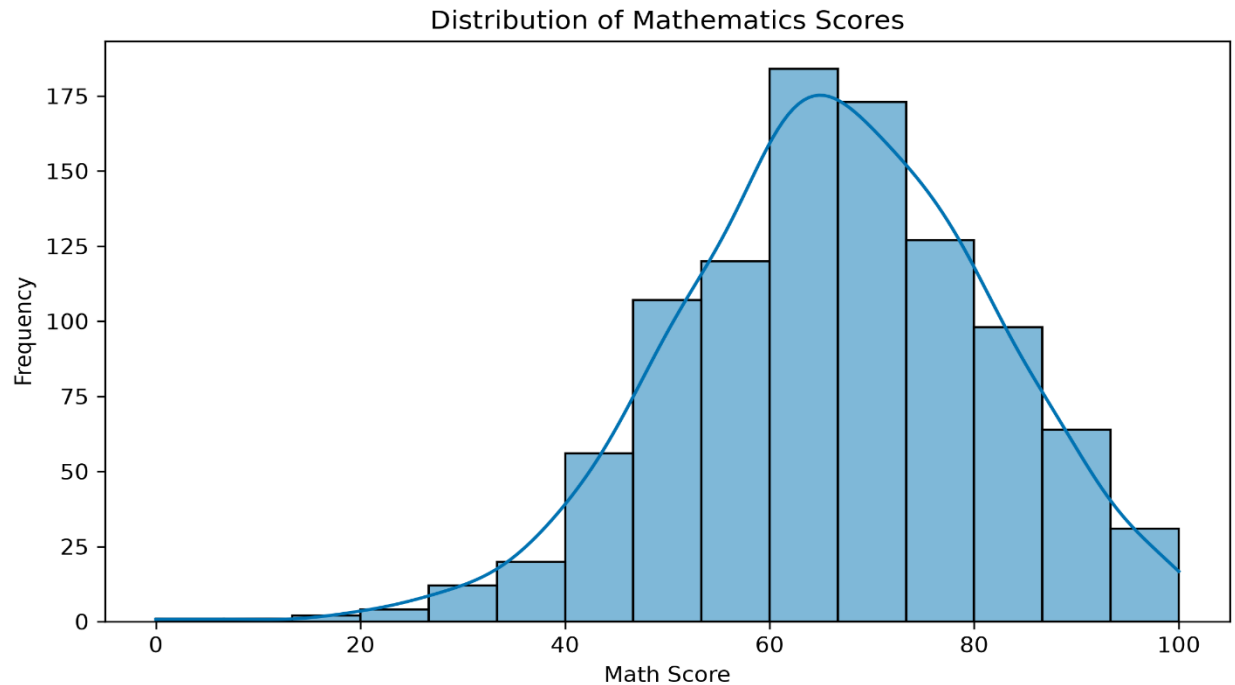
To assess practical significance, eta-squared (η^2) was calculated. The resulting effect size ($\eta^2 = 0.0318$) indicates that approximately 3.18% of the variation in mathematics scores can be explained by parental education level.

Because the analysis was observational and did not control for other explanatory variables (no confounding variable assessment), it is possible that additional factors contributed to the relationship observed between parental education and mathematics performance.

Figures referenced in the notebook:

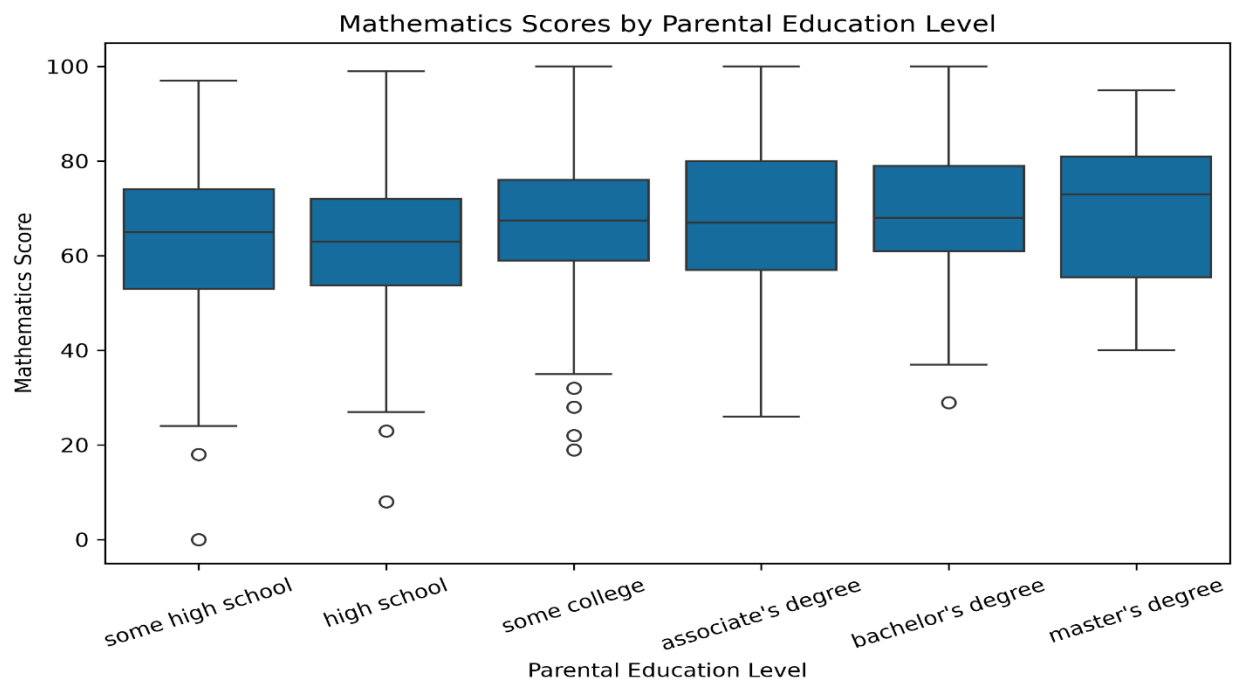
- **Figure 1:** Distribution of Mathematics Scores

The distribution is unimodal and approximately bell-shaped with slight negative skewness, indicating that most students scored near the center of the distribution while a smaller number achieved lower scores. The absence of substantial skewness or multimodality supports the suitability of mathematics scores for subsequent parametric analysis.



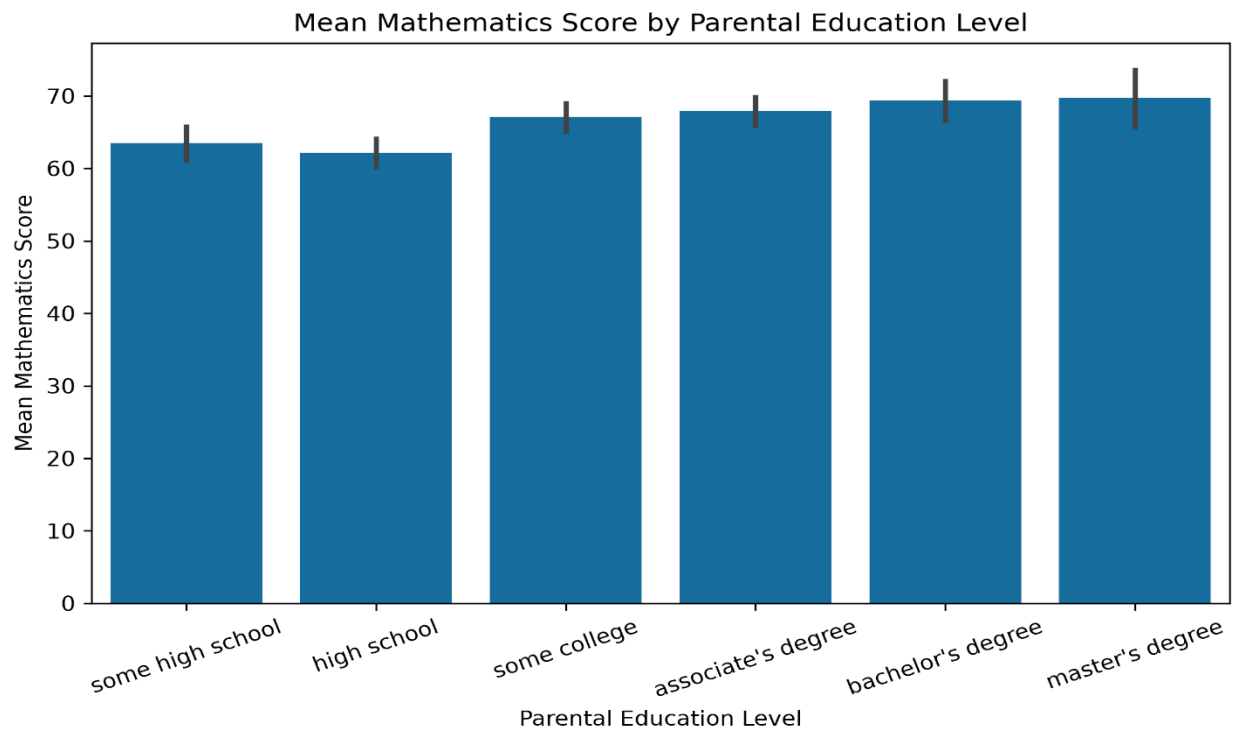
- Figure 2: Mathematics Scores by Parental Education Level**

Students whose parents attained higher levels of education generally exhibited higher median mathematics scores. Variability appears reasonably consistent across groups, and the limited number of outliers suggests that the homogeneity of variance assumption is likely satisfied.



- **Figure 3: Mean Mathematics Score by Parental Education Level**

Mean mathematics scores by parental education level with 95% confidence intervals (error bars). A general positive trend is visible, with higher parental educational attainment associated with higher average mathematics performance. However, overlapping confidence intervals suggest that formal hypothesis testing is required to determine whether observed differences are statistically significant.



Interpretation for a Non-Technical Audience

The analysis suggests that students whose parents have higher levels of education tend to achieve slightly higher mathematics scores on average than students whose parents have lower levels of educational attainment.

Although these differences were statistically significant, parental education explains only a small proportion of the variation in student mathematics performance. This means that parental education appears to be one factor associated with mathematics achievement, but many other influences likely contribute to student outcomes as well.

Factors such as school quality, household income, learning environment, student motivation, access to educational resources etc. were not included in this dataset and may also play important roles in academic success.

Consequently, the findings should not be interpreted as evidence that parental education alone determines mathematics performance. Rather, the results suggest that parental education is associated with achievement while other factors likely contribute to observed differences among students.

Limitations and Potential Bias

Several limitations should be considered when interpreting these findings.

First, the dataset is observational rather than experimental (important for causal inference). Consequently, the analysis can identify associations between parental education and mathematics performance but cannot establish causal relationships.

Second, the dataset does not include several potentially important variables, such as household income, school quality, geographic location, or student motivation. These omitted variables may influence student performance and could partially explain the observed differences across parental education groups.

Third, although parental education exhibited a statistically significant relationship with mathematics performance, the effect size analysis indicated that only a small proportion of score variability was explained by this factor. This suggests that additional influences not captured in the dataset contribute substantially to educational outcomes.

Finally, the dataset represents a specific student population and may not be fully representative of broader educational contexts. As a result, care should be taken when generalizing these findings to other populations or educational systems.

References

Agarwal, A., Prabha, S., & Yadav, R. (2025). Exploratory data analysis (EDA) for banking and finance: Unveiling insights and patterns. *GBS Impact: Journal of Multi Disciplinary Research*. <https://doi.org/10.58419/gbs.v11i1.1112520>

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Students Performance in Exams [Dataset]. Kaggle.
<https://www.kaggle.com/datasets/spscientist/students-performance-in-exams>